Theory of Computation

Closure

Lecture 34

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$\overline{Reg} = Reg$

Closed

Complement.

$\Sigma = \{a, b\}$

1.) $L = \phi$, $\bar{L} = \Sigma^*$

2.) $L = \Sigma^*$, $\bar{L} = \phi$

3.) $L = a.\Sigma^*$, $\bar{L} = b\Sigma^* + \varepsilon$

4.) $L = b\Sigma^*$,

$\bar{L} = a\Sigma^* + \varepsilon$

$\Sigma^* . (a+b)^*$

$\phi \cup \Sigma^* = \Sigma^*$

5.) $L = \{\varepsilon\}$,

$\bar{L} = (a+b)^+$

$L \cup \bar{L} = \Sigma^*$

6) $L = \Sigma^* a \Sigma^*$

$= (a+b)^* a (a+b)^*$

$\bar{L} = b^*$

7) $L = \Sigma^* b \Sigma^*$

$\bar{L} = a^*$

8) $L = a^* b^* \quad \varepsilon$

aba

$\bar{L} = \cancel{b^* a^*} \quad \varepsilon$

$b^+ a^+ \quad \{ba, \ldots\}$

$(a+b)^*$

$L = ____$

$\cup$

$\cancel{a \; \Sigma^* b \; b^* \; \Sigma^* b \; \Sigma^* + \varepsilon}$

$\bar{L}$

b^*

8) $L = \underline{a^* b^*} \rightarrow \{\varepsilon, a, b, ab, bb, aa, \ldots\}$

$\bar{L} = b^+ \cancel{\Sigma^*} a^+$

$\bar{L} = (a+b)^* \boxed{b^+ a^+} \Sigma^*$ Superset

$\boxed{\Sigma^* \underline{ba} \Sigma^*}$

$b^+ a^+ = \{\underline{ba}, \underline{bbaa}\}$

9.) $L = \underline{b^* a^*}$

$\bar{L} = \Sigma^* ab \Sigma^*$

Difference

$\Sigma = \{a\}$ ① $L_1 = \Sigma^*$

$L_2 = \phi$

$\{\ \}$

$L_1 - L_2 = \Sigma^*$

$L_2 - L_1 =$

$L_1 - L_2 = \{\varepsilon, a, aa, aaa, \ldots\} - \{\ \} = \{\varepsilon, a, aa, \ldots\}$

$L_2 - L_1 = \{\ \underline{\ }\ \} - \{\varepsilon, a, aa, \ldots\} = \{\ \}$

a

$Reg_1 - Reg_2 = Reg$

Closed

$\Sigma = \{a, b\}$

2.) $L_1 = a\Sigma^*$

$L_2 = b\Sigma^*$

$L_1 - L_2 = \cancel{\phi}\ L_1$

$L_2 - L_1 = \cancel{\phi} = L_2$

$L_1 = \{a, aa, ab, aaa, aab, aba, abb \ldots\}$

$- \{b, ba, bb, baa, bab, bbb, \ldots\}$

$\therefore$

3.) $L_1 = \Sigma^*$

$L_2 = a\Sigma^*$

$$L_1 - L_2 = b\Sigma^* + \varepsilon$$

$$L_2 - L_1 = \phi$$

$$L_2 = \{a, aa, ab, \ldots\} - \{\varepsilon, a, \ldots\}$$

$$- L_1$$

$$= \phi$$

4) $L_1 = \Sigma^*$

$L_2 = \underline{a^* \cancel{b^*}}$ ε $\{\varepsilon, a, aa, \ldots, b, \ldots, \underline{a} \cdots \underline{b}, \ldots\ldots$

$L_1 - L_2 = \cancel{b^+ a^+}\ \Sigma^* \boxed{b\,a} \Sigma^*$

$L_2 - L_1 = \phi$

Thank You

DTM

$0/1 \longrightarrow 2^n$

$\downarrow$

Non-deterministic Algo $\rightarrow O^n$

NP

$\equiv$

NTM

$\downarrow$

$0/1 \longrightarrow O^n$

$|||$

NP